SanadAI at IslamicEval 2026: Detecting Hallucinations in Arabic Islamic Citations through Span Detection and Hybrid Verification

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Abstract

We present SanadAI, a modular system for the IslamicEval 2026 shared task on fine-grained hallucination detection and correction in Arabic Islamic content. We participate in Subtasks 1 and 2, which address citation span detection and hallucination detection, respectively. For Subtask 1, we fine-tune AraModernBERT as a BIO sequence tagger to identify Quranic verses (*ayah*), Hadith text (*matn*), transmission chains (*isnad*), and claimed sources. Controlled Quranic and Hadith augmentation, together with Shamela-based hard-negative selection, substantially improves citation span detection over training on competition data alone. For Subtask 2, we use source-specific verification pipelines: Quranic evidence is verified through Arabic normalization and approximate matching, while Hadith evidence is verified using BGE-M3 dense retrieval, FAISS candidate search, lexical reranking, and semantic and entity-level evidence. SanadAI achieves 97.35% Macro-F1 on Subtask 1 and 86.66% official macro-accuracy on Subtask 2 on the hidden test set. Quranic verification is strongest at 96.89%, whereas isnad verification remains the most challenging at 77.42%, reflecting the difficulty of verifying transmission chains when the system relies on narrator compatibility with a retrieved Hadith rather than explicitly modeling narrator order and chain continuity.

1 Introduction

Large language models (LLMs) can generate fluent Arabic responses containing fabricated, altered, or incorrectly attributed Islamic citations. Such errors are particularly difficult to detect when the generated text is linguistically natural but the cited evidence is incorrect. IslamicEval addresses this problem in Arabic Islamic content, with the 2026 edition introducing a fine-grained formulation that distinguishes different citation types (Mubarak et al., 2025; Alharbi et al., 2026).

IslamicEval 2026 defines two subtasks that we address: **Subtask 1, citation span detection**, identifies *ayah*, *matn*, *isnad*, and *claimed source* spans; **Subtask 2, hallucination detection**, determines whether the corresponding evidence is supported by the appropriate reference collection. This naturally separates the problem into two stages: identifying *what and where the evidence is* and determining *whether it is valid*.

We present **SanadAI**, a modular system following this decomposition. For Subtask 1, we fine-tune an Arabic encoder for token-level citation extraction using controlled data augmentation and hard-negative examples. For Subtask 2, we employ source-specific verification pipelines. Quranic citations are matched against the compact, fixed Quranic collection, whereas Hadith verification requires retrieval from a substantially larger collection and additional reasoning over matn, source, and isnad information.

The two official submissions are evaluated independently. We nevertheless describe them in a common pipeline because the resulting decomposition provides a clear separation between citation localization and evidence verification.

Our contributions are:

- a BIO-based citation span detector for *ayah*, *matn*, *isnad*, and *claimed source*;
- controlled Quranic and Hadith augmentation with Shamela-based hard-negative examples;
- a source-aware verification framework combining approximate matching, dense retrieval, lexical similarity, and entity-level evidence;
- release of the code, augmented data, and trained model for reproducibility.[1]

[1]Code: https://github.com/AbirKorched/SanadAI-IslamicEval2026; Subtask 1 data: https://huggingface.co/datasets/AbirKorched9/

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2 Background

IslamicEval 2025 introduced hallucination detection and correction for quoted Quranic and Hadith text, with participating systems exploring retrieval, semantic matching, and LLM-based verification (Mubarak et al., 2025; Al Adel et al., 2025; Omayrah et al., 2025). IslamicEval 2026 extends this setting with a finer-grained formulation that explicitly separates citation span identification from evidence verification (Alharbi et al., 2026).

For fine-grained evidence span detection, transformer encoders are well established for sequence labeling (Devlin et al., 2019), while Arabic-specific pretraining improves performance on Arabic language understanding (Antoun et al., 2020). We use AraModernBERT (Elshehy et al., 2026) for span detection and GLiNER (Zaratiana et al., 2024) to extract entities such as narrators and sources during verification.

Verification requires retrieval over source-specific collections. BGE-M3 provides multilingual dense representations (Chen et al., 2024), while FAISS enables efficient similarity search (Douze et al., 2024). We use these components for Hadith retrieval, where the evidence space is substantially larger and more heterogeneous than the compact, fixed Quranic collection. SanadSet 650K and Shamela provide additional Hadith resources for retrieval and controlled augmentation (Mghari et al., 2022; Belinkov et al., 2016).

3 System

SanadAI consists of two complementary components corresponding to the two shared-task subtasks. Figure 1 summarizes the overall organization. Subtask 1 performs citation localization using AraModernBERT, while Subtask 2 verifies the resulting evidence using source-specific matching and retrieval pipelines. The figure shows the conceptual relationship between the components; the official submissions are evaluated independently.

3.1 Subtask 1: Citation Span Detection

We formulate Subtask 1 as BIO sequence classification over four entity types: *ayah*, *matn*, *isnad*, and *claimed source*. We fine-tune `NAMAA-Space/AraModernBert-Base-V1.0` to predict token-level labels using class-weighted cross-entropy to mitigate the strong class imbalance caused by the dominance of the background class.

To encourage the model to learn recurring citation patterns, we augment the official data with Quranic and Hadith examples and manually selected Shamela-based hard negatives. The augmentation exposes the model to recurring citation structures, including longer *ayah* and *matn* spans, *isnad* preceding *matn*, and *claimed source* spans associated with *ayah* or *matn* citations. The 20 Shamela-based hard negatives were manually selected from classical books to resemble Hadith-style narration while containing neither actual Hadith nor Ayah spans. The selected configuration combines the competition data with 6K synthetic Quranic and 6K synthetic Hadith examples; dataset composition and construction are detailed in Appendix D.

During inference, token probabilities are converted to word-level predictions using a confidence threshold of 0.45. Adjacent fragments are merged, while simple validation and boundary cleaning remove implausibly short predictions and surrounding punctuation. The complete training and inference pipeline is provided in Appendix B.

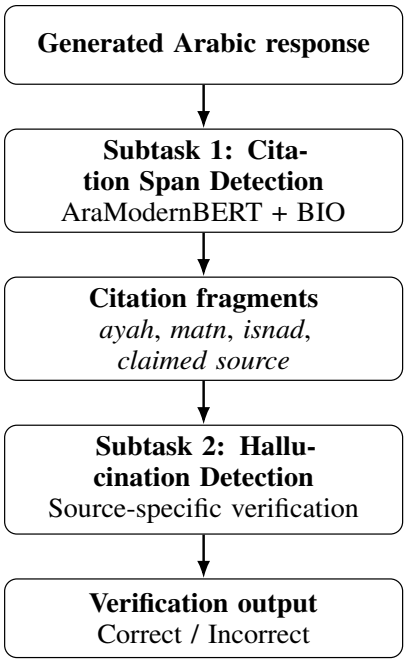


Figure 1: SanadAI overview: citation span detection followed by source-specific evidence verification.

3.2 Subtask 2: Hallucination Detection

We implement Subtask 2 as source-specific evidence verification. Figure 3 summarizes the

`IslamicEval2026-Task1-augmented`; Subtask 2 resources: https://huggingface.co/datasets/AbirKorched9/IslamicEval2026-Task2; model: https://huggingface.co/AbirKorched9/SanadAI-AraModernBERT-IslamicNER

Quranic and Hadith pipelines, which are processed separately because their reference collections and textual structures differ substantially. The complete verification pipelines are provided in Appendix C.

3.2.1 Quranic Verification

For Quranic evidence, generated text and references are normalized to reduce variation in Arabic orthography, diacritics, punctuation, and whitespace. Corpus-wide approximate matching is then performed over the 114 surahs. GLiNER extracts structured source information such as surah and verse identifiers, which is checked against the retrieved reference.

3.2.2 Hadith Verification

For Hadith evidence, the Hadith reference collection provided by the organizers, containing approximately 34K records, is encoded with BGE-M3 and indexed using FAISS. For each generated *matn*, the top-$K = 2048$ candidates are retrieved, followed by RapidFuzz-based lexical selection of the top-1 candidate. The selected candidate is then evaluated using semantic and lexical evidence, while claimed source and isnad verification additionally use GLiNER-based entity matching.

The verification score is generally expressed as

$$S = \alpha S_{\text{sem}} + (1 - \alpha) S_{\text{aux}}, \tag{1}$$

where S_{aux} denotes lexical similarity for Hadith matn verification and entity similarity for metadata verification. Component-specific coefficients and thresholds are selected on the development set; the complete configuration is reported in Appendix A.

4 Results

We evaluate the system using the official IslamicEval scoring procedures. Subtask 1 is evaluated using character-level Macro-F1 over the four citation classes. Subtask 2 uses the official macro-average of per-type accuracies, excluding gold annotations marked *N/A*.

Table 1 summarizes development and hidden-test performance.

4.1 Subtask 1 Results

SanadAI achieves 97.35% Macro-F1 on the hidden test set, compared with 96.61% on development data. The large improvement over training on competition data alone demonstrates the importance of the augmented training set.

Metric / Entity	Dev (%)	Test (%)	Δ
Subtask 1 Macro-F1	96.61	**97.35**	+0.74
Subtask 2 Macro-acc	91.05	**86.66**	-4.39
Ayah	97.42	96.89	-0.53
Matn	94.56	88.56	-6.00
Claimed Source	85.55	83.76	-1.79
Isnad	86.67	77.42	-9.25

Table 1: Official development and hidden-test performance. Δ denotes Test − Dev in percentage points.

Training Configuration	F1 (%)	Δ
Competition data only	81.65	–
+ 2K Quran + 2K Hadith + 20 HNs	95.77	+14.12
+ 6K Quran + 6K Hadith + 20 HNs	**96.41**	**+14.76**
+ 8K Quran + 8K Hadith + 20 HNs	95.78	+14.13
+ Best aug. + post-processing	**96.61**	**+14.96**

Table 2: Subtask 1 data augmentation ablation on the development set. Δ denotes the change in Macro-F1 relative to the competition-only baseline.

Table 2 shows the effect of augmentation size. Training on the competition data alone obtains 81.65% development Macro-F1. Adding 2K + 2K increases this to 95.77%, while 6K + 6K reaches 96.41%. Increasing the augmentation further to 8K + 8K reduces performance to 95.78%, suggesting that additional synthetic data do not necessarily provide further benefit. The final post-processing stage increases the selected configuration to 96.61%.

4.2 Subtask 2 Results

SanadAI achieves 86.66% official macro-accuracy on the hidden test set, compared with 91.05% on development data. As shown in Table 1, Quranic verification is the strongest component at 96.89%, followed by Hadith matn at 88.56%, claimed source at 83.76%, and isnad at 77.42%.

The results highlight increasing difficulty from textual to metadata verification. Quranic verification benefits from a fixed collection and direct matching, whereas Hadith verification must retrieve evidence from a larger, more heterogeneous collection. Our pipeline then verifies metadata against only the top-1 ranked Hadith candidate, which is limiting because the same Hadith may appear with different wording or narrator chains. Thus, top-1 candidate verification can miss valid evidence that would be found among alternative references.

Encoder	Dim.	Dev F1 (%)
BGE-M3	1024	**87.30**
GATE-AraBERT-v1	768	86.08
E5-all-nli-triplet-Matryoshka	384	85.71
Arabic-Triplet-Matryoshka-V2	768	85.46
AraModernBERT-Base-V1.0	768	84.34
GTE-multilingual-base	768	83.74

Table 3: Hadith encoder comparison on the development set.

4.3 Hadith Encoder Comparison

Because Hadith matn verification depends on retrieving a suitable reference, we compare six embedding encoders under the same verification pipeline. As shown in Table 3, BGE-M3 achieves the strongest development performance at 87.30% F1 and is therefore selected for the final system.

5 Discussion

5.1 Retrieval and Verification

As shown in Table 1, verification is easier for Quranic evidence, which is matched against a compact, fixed collection, achieving 96.89% accuracy. Hadith matn verification is more challenging because it requires retrieval from a larger collection.

The current top-1 selection makes retrieval errors irrecoverable and may discard valid variants among other candidates. Expanding the Hadith collection, together with multi-candidate verification or learned reranking, could improve robustness.

5.2 Metadata and Isnad Verification

Claimed-source and isnad verification achieve 83.76% and 77.42%, respectively, below direct textual verification. The current isnad verifier compares extracted narrator entities without modeling their order or chain continuity. Preserving the narrator sequence and comparing positions and adjacent relationships could provide more reliable chain verification.

5.3 Development-to-Test Generalization

Subtask 2 decreases from 91.05% macro-accuracy on development data to 86.66% on the hidden test set, with the largest degradation occurring for isnad verification. This suggests that development-set threshold selection does not fully transfer to unseen examples, particularly for metadata-level verification. More robust calibration or jointly learned verification models could reduce this sensitivity.

6 Conclusion

We presented SanadAI, a lightweight system for IslamicEval 2026 that combines fine-grained citation span detection with source-specific evidence verification. AraModernBERT with controlled augmentation achieves 97.35% Macro-F1 on Subtask 1, while the hybrid Quranic and Hadith pipelines achieve 86.66% official macro-accuracy on Subtask 2. The results show that targeted augmentation substantially improves citation localization, while verification remains limited by retrieval coverage and the structural complexity of Hadith evidence. In particular, isnad verification is constrained by reliance on a single top-ranked candidate and narrator compatibility without explicitly modeling chain order and continuity.

Acknowledgments

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A Complete Experimental Configuration

The complete configuration of the submitted system is summarized in Table 4, including training, retrieval, and verification parameters.

For Hadith matn verification, semantic and lexical similarity are combined as

$$S_{\text{matn}} = \alpha_{\text{matn}} S_{\text{sem}} + (1 - \alpha_{\text{matn}}) S_{\text{lex}}, \quad (2)$$

where S_{sem} is the BGE-M3 similarity and S_{lex} is the normalized lexical similarity.

For claimed-source and isnad verification, semantic similarity is combined with an entity-compatibility score:

$$S_{\text{meta}} = \alpha_{\text{meta}} S_{\text{sem}} + (1 - \alpha_{\text{meta}}) S_{\text{ent}}, \quad (3)$$

where the coefficient α_{meta} is selected separately for claimed-source and isnad verification.

The entity score is computed as

$$S_{\text{ent}} = \frac{1}{|C|} \sum_{c \in C} \max_{t \in T} \text{sim}(c, t), \quad (4)$$

where C is the set of entities extracted from the generated evidence and T is the corresponding set extracted from the reference. The function sim denotes the fuzzy string similarity used by the implementation. If no entities are extracted from the generated evidence, S_{ent} is set to zero.

Each verification component uses a development-selected threshold. An instance is classified as *correct* when its final score is at least the corresponding threshold and as *incorrect* otherwise. For isnad verification, development tuning selected $\alpha_{\text{isnad}} = 0$, so the final score depends entirely on entity compatibility. Thus, the submitted verifier checks narrator compatibility and does not explicitly model narrator order or transmission-chain continuity.

B Subtask 1 Pipeline

Figure 2 shows the training and inference pipeline used for citation span detection.

C Subtask 2 Pipeline

Figure 3 summarizes the source-specific verification pipelines.

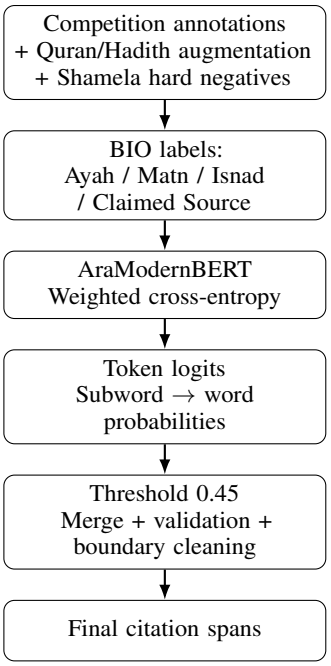


Figure 2: Subtask 1 citation span detection pipeline.

C.1 Quranic Verification

The Quranic reference collection is organized by surah and verse. Generated and reference text are normalized by removing diacritics and annotation marks, normalizing common Arabic character variants, removing tatweel and punctuation artifacts, and collapsing repeated whitespace.

RapidFuzz is then applied corpus-wide over the 114 surahs. For claimed sources, GLiNER extracts surah names, surah numbers, and verse numbers. The extracted metadata are compared with the expected reference, while the corresponding canonical verse is compared with the generated ayah.

C.2 Hadith Matn Verification

The Hadith reference collection contains approximately 34K records. References are encoded using BGE-M3 and indexed with FAISS. For each generated matn, the top 2048 candidates are retrieved. RapidFuzz partial-ratio similarity selects the candidate with the strongest lexical match, after which semantic and lexical scores are combined.

The development-selected configuration uses $\alpha_{\text{matn}} = 0.45$ and a decision threshold of 0.74.

C.3 Hadith Claimed-Source Verification

GLiNER extracts source-related entities from generated and expected source information, including books, scholars, and narrators. Entity similarity is combined with semantic similarity. The selected

Subtask	Component	Configuration
Subtask 1	Encoder	`NAMAA-Space/AraModernBert-Base-V1.0`
	Training objective	Class-weighted cross-entropy
	Epochs	15
	Learning rate	1×10^{-5}
	Maximum sequence length	1024
	Gradient accumulation	16
	Augmentation	6K Quranic + 6K Hadith examples + 20 hard negatives
	Inference	Threshold 0.45; merging and boundary cleaning
Subtask 2: Quran	Matching	RapidFuzz over the 114 surahs
	Source metadata extraction	`NAMAA-Space/gliner_arabic-v2.1` for surah and verse information
	Source verification	Compare extracted metadata with the source of the matched ayah
Subtask 2: Hadith	Encoder	`BAAI/bge-m3`
	Retrieval	FAISS, top-2048 candidates from 34K Hadith records
	Top-1 candidate selection	RapidFuzz partial-ratio
	Matn verification	$\alpha = 0.45$, threshold $= 0.74$
	Source metadata extraction	`NAMAA-Space/gliner_arabic-v2.1` for narrator extraction
	Source verification	$\alpha = 0.45$, threshold $= 0.17$
	Isnad verification	$\alpha = 0.00$, threshold $= 0.70$

Table 4: Configuration of the submitted SanadAI system.

configuration uses $\alpha_{\text{source}} = 0.45$ and a threshold of 0.17.

C.4 Hadith Isnad Verification

GLiNER extracts narrator entities from the generated and reference isnads. The verifier computes narrator compatibility using fuzzy entity similarity and combines this score with semantic similarity. Development optimization selects $\alpha_{\text{isnad}} = 0.00$ and a threshold of 0.70.

The final verifier therefore relies entirely on entity compatibility. It does not explicitly model narrator ordering or chain continuity, and consequently verifies narrator compatibility rather than complete transmission-chain validity.

D Subtask 1 Data Construction

To improve span detection performance, we construct an augmented training dataset by combining the official shared-task training set with synthetic data and targeted hard negatives.

D.1 Quranic Augmentation

We generate 6,000 Quranic examples where each instance contains a single ayah or consecutive ayahs. These are preceded by varied introductory phrases and quotation styles, and paired with a claimed-source statement sampled from multiple templates. We also introduce Arabic and Eastern Arabic numeral variants.

D.2 Hadith Augmentation

We curate 6,000 unique Hadith instances from the SanadSet 650K corpus (Mghari et al., 2022). Each example is constructed from a single Hadith by extracting its *isnad* and *matn*, applying standard Arabic text normalization, and synthesizing claimed-source statements through diverse linguistic templates.

D.3 Shamela Hard Negatives

To expose the span detector to citation-like text that should not be classified as a citation, we include 20 manually selected passages from the Shamela library. These passages exhibit Hadith-like narrative cues but contain neither actual Hadith nor Ayah spans, thereby providing challenging negative examples to reduce citation-span false positives.

D.4 Final Training Dataset

The final training dataset comprises 4,706 official shared-task instances, 6,000 Quranic examples, 6,000 Hadith examples, and 20 Shamela hard negatives, totaling 16,726 instances. The synthetic generation pipelines and reproduction scripts are available online.[2]

[2] https://huggingface.co/datasets/AbirKorched9/IslamicEval2026-Task1-augmented

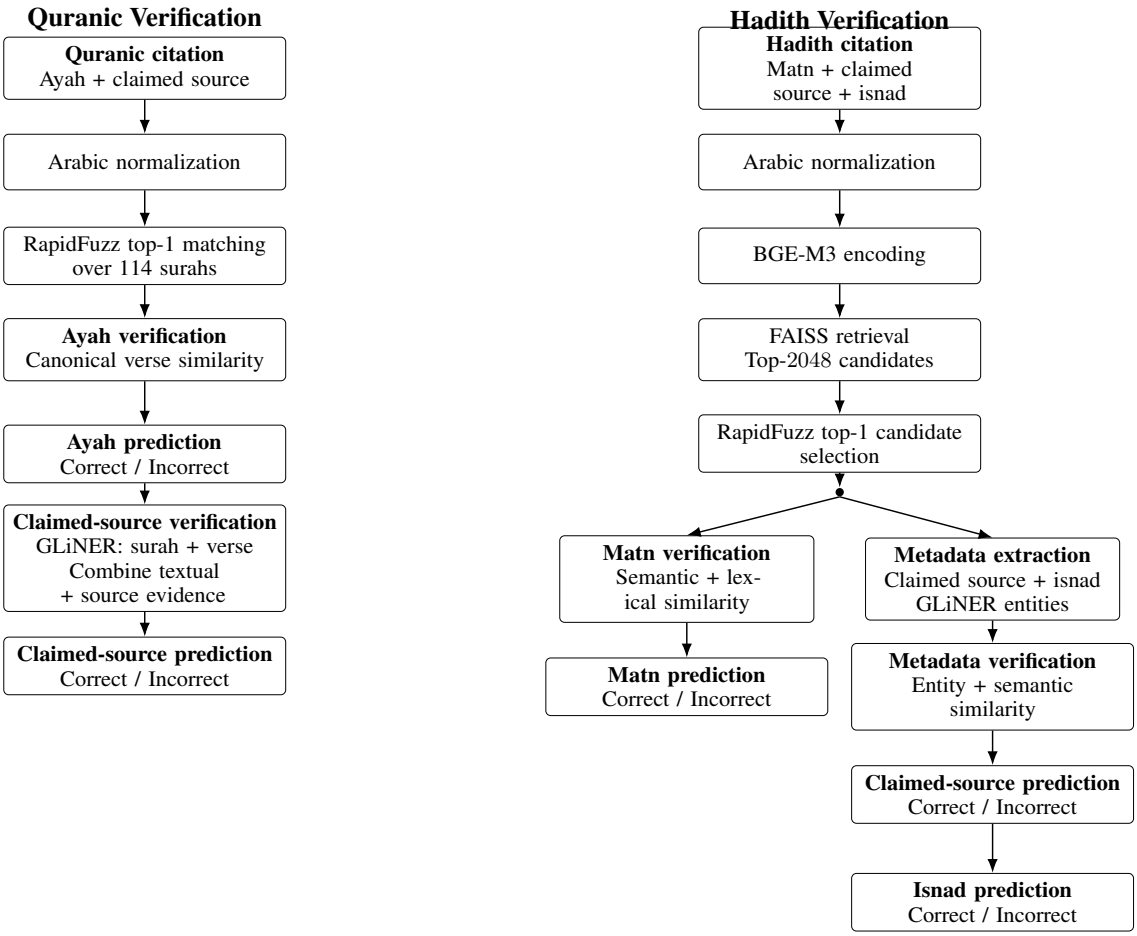


Figure 3: Subtask 2 source-specific verification pipelines. The Quranic pipeline normalizes the citation, retrieves the most similar canonical verse, and independently verifies the Ayah and claimed source. The Hadith pipeline retrieves candidate references using BGE-M3 and FAISS, selects the top candidate using lexical similarity, and verifies the Matn, claimed source, and isnad using semantic, lexical, and entity-level evidence.